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About the earthquake

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Why do earthquakes happen?

An earthquake refers to a phenomenon in which the bedrock in the underground is pushed or pulled from the surroundings, and the bedrock suddenly shifts with a certain surface as a boundary. The shaking (seismic waves) caused by the sudden devision of this bedrock is transmitted to the surroundings, and when it reaches the surface of the earth, the earth's surface "shakes". We know that this "shaking" caused the earthquake to occur underground.

What is the epicenter area?

Earthquakes are caused by the misalignment of the underground bedrock. The area where the bedrock shift (fault) occurred when an earthquake occurred is called the epicenter area. In general, the length of the epicenter is about a few kilometers in a magnitude 5 earthquake, about tens of kilometers in a magnitude 7 earthquake, about 100 to 200 km in a magnitude 8 earthquake, and about 500 to 1000 km in a magnitude 9 earthquake. Is. In addition, the epicenter refers to the place where the misalignment of the bedrock began, while the epicenter refers to the entire area where the misalignment of the bedrock occurred.

What is a fault?

An earthquake refers to a phenomenon in which the bedrock in the underground is pushed or pulled from the surroundings, and the bedrock suddenly shifts with a certain surface as a boundary. This shift is called fault. It is called this way to cut off the surrounding strata due to the misalignment of the bedrock caused by the earthquake. The fault has a surface spread, which is also called a fault surface. As the depth of the epicenter approaches the surface, faults may appear to the surface of the earth, and examples of such faults include the Nojima Fault on Awaji Island in Hyogo Prefecture and the Neodani Fault in Motosu City, Gifu Prefecture.

What is an active fault? What kind of active faults are there in ○○ city, and how likely are earthquakes?

Among the topography that is usually recognized as faults on the surface of the earth, those that are active in the recent geological era (in this case, it refers to roughly 1.7 to 2 million years ago in the Quaternary period to the present) and are likely to be active in the future (= causing earthquakes) It is called an active fault. The earthquake investigation committee of the government's earthquake survey and research promotion headquarters conducts an evaluation of the possibility of earthquakes for each active fault. Please see the following page for details.

Related links

- Earthquake Research Promotion Headquarters

What is a plate?

Plate (plate) originally means board in English. In the field of geology, it is a layer of rock that covers the earth's surface. Compared to the radius of the Earth, which is about 6,400 km, the plate is called this because it looks like a plate about 10 to 100 km thick. If you look at the places where earthquakes occur around the world, they are arranged in a long and strip. This band corresponds to the boundary between plates, and the world is covered with more than a dove of major plates without gaps. Each plate is moving relatively, and at its boundaries, the plates collide with each other, or the other plate sneaks (sinks) under one plate. At that time, an earthquake occurs due to the force applied to the plate.

Related links

- The place where the earthquake occurs

What is an aftersheck? Why do you know that an earthquake that occurred more than a month later is an aftershock?

After a major earthquake, seismic activity may be active near the epicenter. After a major earthquake, earthquakes smaller than the first earthquake often occur near the epicenter, which is called aftershocks, and the largest earthquake that first occurred is called a main earthquake. However, in some cases, an earthquake larger than the first earthquake may occur, in which case it will be the main earthquake, and the earthquake that occurred before it may be called a foremor earthquake.

Afterstocks occur more often immediately after a major earthquake, and they gradually decrease over time, but some last for more than a month or even decades. Aftershocks are believed to occur in the process of resolving the unstable state of underground force balance caused by the occurrence of the main earthquake. In general, it is common to consider the place where many earthquakes occur along the main earthquake fault as the aftershock area, and treat the earthquakes that occur in it as aftershocks, but it is impossible to clearly distinguish whether the earthquake that occurs is an aftershock or not.

Related links

- About seismic activity (aftershocks, etc.) after a major earthquake

What is the blank area of an earthquake?

If you draw a diagram of earthquake distribution, there may be places where there is seismic activity in the surrounding area, but only that part does not occur (or is relatively quiet). This is called a blank space. There are two main types of blank space.

If you draw the epicenter of a trench-type earthquake on the map, each of them does not overlap and has the property of lining up along the trench, but gaps may be seen between these epicenters. Such a place is considered to be a place where there is a potential for a major earthquake, although there has been no major earthquake for a long time recently, and this is called a type 1 blank area.

Major earthquakes that cause damage are rare, but smaller earthquakes occur on a daily basis, including tiny earthquakes that the human body does not feel. The number of such daily earthquakes may temporarily decrease in an area, and then a major earthquake may occur in that area. This phenomenon of a decrease in the number of earthquake occurrences on a daily basis is called the calming of seismic activity, and the area where the phenomenon appears is called the second-class blank area.

In this way, not all areas where earthquakes have not occurred are called earthquake blank areas.

What kind of earthquake is a direct earthquake?

In general, "directly below earthquakes" are earthquakes that occur directly below urban areas, etc., and it seems that they often refer to those that cause great damage, but there is no clear seismological definition of "directly below earthquakes".

The scale of shallow earthquakes that occur on land is often smaller than that of huge earthquakes that occur near the trent, but because the place where the earthquake occurs is shallow, it is easy to shake more directly above, and if people live there, the magnitude is 6~ Even about 7 can cause great damage.

How many earthquakes have occurred in the world and around Japan?

The number of earthquakes occurring in the world on average per year is shown in Table 1.

Table 1 Number of earthquakes in the world (average per year: according to USGS (United States Geological Survey))

Magnitude	Number of times (average for 1 year)	Remarks
M8.0 or higher	1	Based on data from 1900 to 2021
M7.0 - 7.9	14	According to data from 1990 to 2021
M6.0 - 6.9	136	According to data from 1990 to 2021
M5.0 - 5.9	1,524	According to data from 1990 to 2021

In addition, the average number of earthquakes occurring in Japan and its surroundings in a year is shown in Table 2. If you compare the number of earthquakes at magnitude 5 or more with Table 1, you can see that the number of earthquakes in Japan and its surroundings is almost 1/10 of the earthquakes in the world.

Table 2 Number of earthquakes in and around Japan (average for 1 year)

※It is calculated based on the epicenter data of the Japan Meteorological Agency from 2014 to 2023.

Magnitude	Number of times (average for 1 year)
M8.0 or higher	0.1 (Once every 10 years)
M7.0 - 7.9	1
M6.0 - 6.9	17
M5.0 - 5.9	134
M4.0 - 4.9	About 1,000

In addition, the number of earthquakes with a magnitude of 5.0 or higher that occurred in and around Japan in 2011 is shown in Table 3. Due to the extremely active aftershock activity after the "Heisei 23 (2011) Tohoku Pacific Ocean Earthquake" on March 11, 2011, the number far exceeded the annual average shown in Table 2.

Table 3 Number of earthquakes in and around Japan (2011)

※It is calculated based on the epicenter data of the Japan Meteorological Agency

Magnitude	Number of times (2011)
M8.0 or higher	1
M7.0 - 7.9	8
M6.0 - 6.9	107
M5.0 - 5.9	691

What is the largest earthquake in the world?

The magnitude of the earthquake is expressed in magnitude (M). The magnitude usually used in earthquake information is only the maximum amplitude of the seismic wave, but in general, when it comes to large-scale earthquakes, the maximum amplitude gradually increases according to the scale (magnitude saturation). In order to avoid such saturation, in large-scale earthquakes, Mw (moment magnitude), which is a magnitude using information on the period and amplitude of seismic waves, may be used. The largest earthquake in the world is the Mw 9.5 earthquake that occurred in Chile, South America on May 22, 1960, since 1900 AD. The epicenter of this earthquake is as long as 1,000 km. In addition, the tsunami hit Japan across the Pacific Ocean for about a day, causing great damage. The large-scale earthquakes that have occurred since 1900 by the U.S. Geological Survey are as follows. (As of August 1, 2025. However, the Mw of the "Heisei 23 (2011) Tohoku Pacific Ocean Earthquake" is according to the Japan Meteorological Agency.)

Top 10 earthquakes with the largest scale that have occurred since 1900

Ranking	Date and time (Japan time)	Place of occurrence	Magnitude (Mw)
1	May 23, 1960	Chile	9.5
2	March 28, 1964	Gulf of Alaska	9.2
3	December 26, 2004	Indonesia, northern western coast of Sumatra	9.1
4	March 11, 2011	Off the coast of Sanriku, Japan "Heisei 23 (2011) Tohoku Pacific Ocean Earthquake"	9.0
	November 5, 1952	Kamchatka Peninsula	9.0
6	February 27, 2010	Off the coast of Chile and Mauri	8.8
	February 1, 1906	Off the coast of Ecuador	8.8
	July 30, 2025	Kamchatka Peninsula	8.8
9	February 4, 1965	Alaska, Aleutian Islands	8.7
10	August 15, 1950	Tibet, Assam	8.6
	April 11, 2012	Indonesia, northern western coast of Sumatra	8.6
	March 29, 2005	Northern Sumatra, Indonesia	8.6
	March 9, 1957	Alaska, Aleutian Islands	8.6
	April 1, 1946	Alaska, Aleutian Islands	8.6

What is the largest earthquake in Japan?

Compared to the moment magnitude (Mw), after 1900, it is the "2011 Tohoku Pacific Ocean Earthquake" (Mw9.0) that occurred on March 11, 2011.

Is there any place in Japan where earthquakes do not occur?

There is no place in Japan where earthquakes do not occur. Small-scale earthquakes occur anywhere in Japan. In addition, even if a large-scale earthquake has occurred in a certain place in the past, there may be no traces (such as active faults) left on the surface of the earth. For this reason, even if there are no active faults or plate boundaries nearby, there is no place where you can say that "there will be absolutely no large-scale earthquakes in this place".

What should I watch out for during an earthquake?

If you see or hear an emergency earthquake warning, or feel the shaking of the earthquake, don't be in a hurry, and first ensure your personal safety. Specifically, protect your head, stay away from large furniture, move to a space (safe space) where things may not fall or fall from around you, or hide under a sturdy desk to ensure your personal safety. Don't be in a hurry to extinguish the fire after the shaking subsides.

After the shaking subsides, we will extinguish the fire, check the information about the earthquake on TV, radio, etc., talk to the neighbors, and evacuate. Evacuate on foot and keep your luggage to a minimum.

In addition, if you feel strong shaking or long and slow shaking near the coast, a tsunami may hit immediately, so it is important to evacuate to a high ground as soon as possible without waiting for the announcement of a tsunami warning or tsunami warning.

To protect yourself from earthquakes, it is very important to be prepared in advance. From normal times, fix the earthquake resistance of the furniture, reinforce the earthquake resistance of the building, secure a safe space, prepare emergency items, and check the evacuation site.

I heard that there will be a big earthquake in the Kanto region soon, what is the situation? What is the situation in other regions?

Trench-type earthquakes and active faults across the country are evaluated by the Earthquake Investigation Committee of the Government's Earthquake Research Promotion Headquarters.

In the Kanto region, the Taisho Kanto earthquake of September 1, 1923 and the Genroku Kanto earthquake of 1703 occurred. These earthquakes are both M (magnitude) class 8 trench-type earthquakes, which are thought to occur at intervals of 180 to 590 years. Since the latest earthquake was the 1923 Kanto earthquake (the Great Kanto Earthquake), it is believed that this type of earthquake is not yet urgent.

During the M8 class earthquakes such as the Taisho Kanto earthquake, there were several M7 class earthquakes, and between the Genroku Kanto earthquake and the Taisho Kanto earthquake, the Kaei Odawara earthquake (1853), the Ansei Edo earthquake (1855), and the Meiji Tokyo earthquake (189 4 years) etc. have occurred. These earthquakes are also causing damage.

In the long-term evaluation of earthquakes along the Sagami Trough by the Earthquake Investigation Committee of the Earthquake Research Promotion Headquarters (2nd edition), the probability of these M7-class earthquakes occurring in the southern Kanto region is an average of 27.5 years in the 220 years between the Genroku Kanto earthquake and the Taisho Kanto earthquake. Since the frequency of earthquakes of about M7 has occurred once, it is said to be about 70% in the next 30 years, and the Central Disaster Prevention Council also considered the estimate and countermeasures of damage assuming the earthquake directly under the capital.

Related links

- Earthquake Survey and Research Promotion Headquarters Long-term evaluation of trench-type earthquakes
- Earthquake Survey and Research Promotion Headquarters Long-term evaluation

What was the name of the earthquake that hit Kobe City on January 17, 1995?

The name of the earthquake was determined by the Japan Meteorological Agency as "1995 Southern Hyogo Prefecture Earthquake". In response to the disaster caused by the "1995 Hyogo Prefecture Southern Earthquake", the government named it the "Great Hanshin-Awaji Earthquake".

Is the "2011 Tohoku Pacific Ocean Earthquake" on March 11, 2011 the same as the "Great East Japan Earthquake"?

No. " The "2011 Tohoku Pacific Ocean Earthquake" is the name of the earthquake set by the Japan Meteorological Agency. " The "Great East Japan Earthquake" is the name of the disaster named by the government for the disaster caused by this earthquake.

Please tell me the criteria for determining the name of the earthquake.

In the event of a significant major earthquake or heavy rain, the Japan Meteorological Agency will determine the name to facilitate emergency and recovery activities after a disaster by disaster prevention-related organizations, etc., and to pass on the experience and valuable lessons of the disaster to future generations. I'm looking forward to it.

For earthquakes, the name is determined by using "era (A.D. year)" and "depitral place name" to make it easier to understand with multiple approximate estimates according to the scale of the earthquake and damage. For more information, please refer to the names of the natural phenomena that caused significant disasters.

In addition, the Japan Meteorological Agency has set a name for the "earthquake" that occurred, but the government may give a different name to the "disaster" caused by the earthquake. For example, the disaster caused by the "1995 Southern Hyogo Prefecture Earthquake" named by the Japan Meteorological Agency is the "Great Hanshin-Awaji Earthquake" as the government, and the disaster caused by the Tohoku Pacific Ocean Earthquake in 2011 is "East Japan Great Earthquake" as the government. It is used to refer to earthquakes and disasters, respectively, such as "disaster".

What is the deepest earthquake in the world?

Depending on the earthquake database (seismic catalog), the USGS (United States Geological Survey) catalog records earthquakes with a depth of more than 700 km near Fiji in the South Pacific.

What is the deepest earthquake in Japan?

In the Japan Meteorological Agency Epicenter Catalog, the aftershock (21:46 on the same day, M3.6) of the earthquake (M8.1, maximum seismic intensity 5 over, depth 682 km) west of the Ogasawara Islands at 20:23 on May 30, 2015 recorded a depth of 698 km.

Have there been earthquakes with a magnitude of 1 or higher in all 47 prefectures nationwide?

In the deep earthquake (M8.1, maximum seismic intensity 5+) off the west coast of the Ogasawara Islands on May 30, 2015, a seismic intensity of 1 or more was observed in all 47 prefectures nationwide. This is the first time since the start of seismic intensity observation in 1885 (Meiji 18).

There was a big earthquake, but is there any dig that it will trigger another big earthquake or volcanic eruption?

In general, it is not clear how one seismic activity affects another earthquake or volcanic activity that occurs in a remote location. The Japan Meteorological Agency monitors the activity of earthquakes and volcanoes 24 hours a day, and announces the activity status in the event of earthquake information and eruption warnings. Please recheck these latest information and your daily preparations for earthquakes and volcanic eruptions.

In addition, it is not possible to predict another major earthquake or volcanic eruption from a large earthquake that occurred, but in the following cases, for example, it may be possible to announce the forecast of seismic activity.

- In and around the epicenter of a major earthquake, the state of the underground forces is unstable, and earthquakes often occur in order to resolve it. The Japan Meteorological Agency will explain the forecast of future seismic activity in the press release materials if there is a possibility of damage from ongoing seismic activity, such as when a maximum seismic intensity of less than 5 is observed in an earthquake.
- In the event of a large-scale earthquake along the Nankai Trough, it may be considered that the possibility of a large-scale earthquake along the Nankai Trough is relatively higher than in normal times. In this case, we will notify you by "Nankai Trough Earthquake Temporary Information".
- In the event of a large-scale earthquake along the Japan Trench and the Chishima Trench, it may be considered that the possibility of large-scale earthquakes along the Japan Trench and the Chishima Trench is relatively higher than in normal times. In this case, we will inform you by the "Hokkaido Sanriku Offshore Post-Earthquake Warning Information".

What are foreshocks, main earthquakes, and aftershocks?

In general, in a series of seismic activities, earthquakes that occur before the largest earthquake are called "front earthquakes", the largest earthquakes are called "main earthquakes", and earthquakes that occur after the main earthquakes are called "aftershocks".

How long does it take for aftershocks (seismological activity after a large earthquake) to subside?

The number of aftershocks decreases over